

Notes: Lauermann & Wolinsky (RES, 2026)

Auctions with Frictions: Recruitment, Entry, and Limited Commitment

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1 Big picture

- **Question.** What happens to auction participation, revenue, and efficiency when three realistic frictions are present simultaneously: costly bidder recruitment, costly bidder entry, and limited seller commitment?
- **Environment.** A seller recruits bidders at a cost, contacted bidders decide whether to pay an entry cost to learn their private value, and then the auction is run. The paper focuses on informal auction settings in which the seller cannot commit to the realized level of participation or disclose it credibly.
- **Core tension.** The seller wants to recruit more bidders because more participation raises the chance of meeting a high-value bidder and, in some formats, intensifies bidding. Bidders, however, dislike paying an entry cost only to face too much competition.
- **Two scenarios.** The paper compares participation-observable auctions (PO), in which bidders observe the realized number of participants before bidding, with participation-unobservable auctions (PU), in which they do not.
- **Main result 1.** In the PO scenario, the stronger recruitment incentives can generate excessive recruitment effort and even market shutdown when the seller cannot commit to moderate participation.
- **Main result 2.** In the PU scenario, there is always a no-trade equilibrium. Because bidders cannot observe participation, the seller has an incentive to secretly under-recruit, which can make expectations unravel and sustain multiple equilibria.
- **Main result 3.** Observability can either promote or suppress trade. When entry costs are low and recruitment costs are high, PO tends to dominate because observed competition raises bids. When entry costs are high and recruitment costs are low, PU may dominate because weaker recruitment incentives reassure bidders and induce entry.
- **Main theoretical contribution.** The paper unifies the literatures on costly recruitment and costly entry, and shows that their interaction with limited commitment creates intrinsic inefficiencies and strong design implications.
- **Main welfare lesson.** Equilibrium generally fails to maximize welfare. Both too much and too little participation can arise, and there can be wasteful recruitment effort whenever entry probabilities adjust downward to offset aggressive recruiting.
- **Why we care.** The model speaks directly to informal auctions, takeover processes, house repairs, and other decentralized price-formation settings where disclosure and commitment are weak. It also helps explain why costly intermediaries such as auction houses may be valuable.

2 Data and measurement (key objects)

2.1 Primitive environment

- A seller offers one indivisible object with value zero to herself.
- The seller chooses recruitment effort $\gamma \geq 0$. The number of contacted bidders is Poisson with mean γ .
- Recruitment cost is linear: γs , where $s > 0$ is the marginal recruitment cost.

- A contacted bidder pays entry cost $c > 0$ to participate and learn his private value.
- Values are i.i.d. from CDF G on $[0, 1]$, with density g and increasing virtual value.
- The paper mostly works with

$$\lambda = q\gamma,$$

the expected number of *actual* participants, where q is the entry probability of a contacted bidder.

2.2 Auction scenarios

- **PO scenario (participation observable).** Bidders see the realized number of participants before bidding.
- **PU scenario (participation unobservable).** Bidders do not see the realized number of participants, and the seller cannot credibly disclose it.
- In both cases, the baseline auction format is a first-price auction without a reserve price.

2.3 Key endogenous objects

- $U(\lambda)$: bidder's ex-ante expected payoff from entering (gross of entry cost), given expected participation λ .
- $R_o(\lambda)$: seller revenue in the PO scenario at participation level λ .
- $R_u(\lambda, \beta_{\hat{\lambda}})$: seller revenue in the PU scenario when actual participation is λ but bidders behave as if expected participation were $\hat{\lambda}$.
- $\bar{\lambda}_c$: maximal participation level that bidders are willing to tolerate, defined by

$$U(\bar{\lambda}_c) = c.$$

- $\bar{s}_o$: maximal marginal recruitment cost for which profitable trade can be sustained in the PO scenario,

$$\bar{s}_o = \max_{\lambda} \frac{R_o(\lambda)}{\lambda}.$$

- $\xi(\lambda)$: in the PU scenario, the marginal revenue of recruiting at actual participation λ when bidder beliefs are correct, i.e.

$$\xi(\lambda) = \left. \frac{\partial}{\partial \lambda} R_u(\lambda, \beta_{\hat{\lambda}}) \right|_{\hat{\lambda}=\lambda}.$$

2.4 Equilibrium notation

- $\lambda_o(s/q)$: the larger solution to $R'_o(\lambda) = s/q$ in the PO case; it is the seller's interior profit-maximizing recruitment scale when it exists.
- $\underline{\lambda}_u(s)$ and $\bar{\lambda}_u(s)$: the smaller and larger solutions to $\xi(\lambda) = s$ in the PU case.
- λ^w and q^w : welfare-maximizing participation and entry magnitudes.

3 Models and empirical specifications (all equations + interpretation)

3.1 Bidding in the PO scenario

Once the number of participants n is realized, the auction is a standard symmetric FPA. The unique equilibrium bid function is

$$\beta_{\text{FPA}}(v, n) = v - \int_0^v \left(\frac{G(y)}{G(v)} \right)^{n-1} dy. \quad (1)$$

- **Interpretation.** A bidder shades below value because he pays his own bid when he wins.
- **Comparative statics.** $\beta_{\text{FPA}}(v, n)$ rises with both v and n . More rivals mean more aggressive bidding.
- **Economic importance.** This is the direct channel through which observability strengthens recruitment incentives in the PO scenario.

3.2 Bidder entry and the break-even participation level

Given beliefs μ over expected participation, a contacted bidder chooses an entry probability q according to

$$\mathbb{E}_\mu[U(\lambda)] > c \Rightarrow q = 1, \quad \mathbb{E}_\mu[U(\lambda)] < c \Rightarrow q = 0. \quad (2)$$

The maximal participation level acceptable to bidders is defined implicitly by

$$U(\bar{\lambda}_c) = c. \quad (3)$$

In the PO scenario, any equilibrium with trade must satisfy

$$\lambda^* \leq \bar{\lambda}_c. \quad (4)$$

- **Interpretation.** If expected participation exceeds $\bar{\lambda}_c$, bidders refuse to enter because the surplus from participation is below the entry cost.
- **Economic intuition.** This is the central entry-side force that disciplines the seller's recruitment decision.

3.3 The bidder payoff object

The appendix gives the ex-ante utility function as

$$U(\lambda) = \int_0^1 \left(\int_0^v e^{-\lambda(1-G(x))} dx \right) g(v) dv. \quad (22)$$

- **Properties.** $U(\lambda)$ is continuous, strictly decreasing, $U(0) = \mathbb{E}[v]$, and $U(\lambda) \rightarrow 0$ as $\lambda \rightarrow \infty$.
- **Interpretation.** Expected participation depresses entry value because winning becomes less likely and surplus conditional on winning becomes smaller.

3.4 Seller profit maximization in the PO scenario

Seller profit is

$$\Pi_o(\lambda, q) = R_o(\lambda) - \lambda \frac{s}{q}.$$

When an interior optimum exists, the seller's necessary conditions are

$$R'_o(\lambda) = \frac{s}{q} \quad \text{and} \quad R''_o(\lambda) \leq 0. \quad (5)$$

This yields the piecewise characterization

$$\arg \max_{\lambda} \Pi_o(\lambda, q) = \begin{cases} \lambda_o(s/q) & \text{if } s/q < \bar{s}_o, \\ \{0, \lambda_o\} & \text{if } s/q = \bar{s}_o, \\ 0 & \text{if } s/q > \bar{s}_o. \end{cases} \quad (6)$$

Therefore, the PO equilibrium participation level satisfies

$$\lambda^* = \begin{cases} \lambda_o(s/q^*) & \text{if } s/q^* < \bar{s}_o, \\ 0 \text{ or } \lambda_o & \text{if } s/q^* = \bar{s}_o, \\ 0 & \text{if } s/q^* > \bar{s}_o. \end{cases} \quad (7)$$

- **Interpretation.** The seller's effective marginal recruitment cost is s/q . When bidders are reluctant to enter, the seller must recruit more intensively to get the same expected participation.
- **Key implication.** The PO scenario has a *minimum viable trade scale* λ_o . If bidders will not tolerate participation that high, the market unravels even if trade could in principle create surplus.

3.5 Revenue in the PO scenario

The seller revenue function is

$$R_o(\lambda) = \int_0^1 \left[1 - e^{-(1-G(v))\lambda} - e^{-(1-G(v))\lambda} (1 - G(v)) \lambda \right] dv. \quad (23)$$

Its derivative is

$$R'_o(\lambda) = \int_0^1 \lambda (1 - G(v))^2 e^{-(1-G(v))\lambda} dv. \quad (24)$$

- **Properties.** R_o is strictly increasing, $R_o(0) = 0$, $R_o(\lambda) \rightarrow 1$, and R'_o is single-peaked.
- **Interpretation.** Recruitment raises revenue by increasing the chance of attracting a high-value bidder and, in the PO case, by increasing the expected transaction price through stronger bidding.

3.6 Observable-participation equilibrium regions

The PO equilibrium is unique (except on knife-edge boundaries) and is summarized in Proposition 1:

$$\text{If } (c, s) \text{ is in Region I, } \lambda^* = \lambda_o(s), q^* = 1; \quad \text{if in Region II, } \lambda^* = \bar{\lambda}_c, q^* = \frac{s}{R'_o(\bar{\lambda}_c)}. \quad (\text{P1})$$

with no trade in the unshaded region of Figure 4.

- **Region I.** Unconstrained trade: bidders are willing to enter at the seller's preferred participation scale.
- **Region II.** Entry-constrained trade: the seller would like to recruit more, but bidders only tolerate $\bar{\lambda}_c$; equilibrium requires $q^* < 1$.
- **Unshaded region.** No trade either because recruitment is too costly or because the seller cannot credibly commit to a participation level acceptable to bidders.

3.7 Commitment benchmark

If the seller could commit to recruitment, she would solve

$$\max_{\lambda} \Pi_o(\lambda, 1) \quad \text{s.t. } \lambda \leq \bar{\lambda}_c.$$

- **Interpretation.** Commitment removes the wasteful need to over-recruit when bidders would otherwise scale entry down.
- **Result.** Commitment is valuable only in Regions II and III of Figure 5: either because equilibrium without commitment wastes recruitment effort, or because commitment can rescue trade that would otherwise unravel.

3.8 Bidding in the PU scenario

When participation is unobservable, bidders condition on expected participation $\hat{\lambda}$ rather than realized participation. The unique symmetric FPA equilibrium is

$$\beta_{\lambda}(v) = \frac{\sum_{n=0}^{\infty} P_{\lambda}(n) G(v)^n \beta_{\text{FPA}}(v, n+1)}{\sum_{n=0}^{\infty} P_{\lambda}(n) G(v)^n}, \quad (11)$$

where $P_{\lambda}(n) = e^{-\lambda} \lambda^n / n!$ is the Poisson probability of facing n competitors.

- **Interpretation.** With unobservable participation, bids reflect beliefs about competition rather than the actual number of rivals.
- **Economic importance.** This disconnect between actual and expected participation is what opens the door to multiple equilibria and no-trade traps.

3.9 Payoff equivalence between PO and PU

Let the expected payoff of a bidder of type v in the PU scenario be

$$u(v, \lambda) = \sum_{n=0}^{\infty} P_{\lambda}(n) G(v)^n (v - \beta_{\lambda}(v)) = \sum_{n=0}^{\infty} P_{\lambda}(n) G(v)^n (v - \beta_{\text{FPA}}(v, n+1)). \quad (12)$$

Ex-ante utility is

$$U(\lambda) = \mathbb{E}_v[u(v, \lambda)]. \quad (13)$$

When beliefs are correct, revenue equivalence implies

$$R_u(\lambda, \beta_\lambda) = \text{Total Surplus}(\lambda) - \lambda U(\lambda) = R_o(\lambda). \quad (14)$$

- **Interpretation.** For any given correctly anticipated participation level, the two scenarios deliver the same bidder payoffs and the same revenue.
- **But.** The equilibrium *participation level* need not be the same, because observability changes the marginal recruitment incentive.

3.10 Seller optimization in the PU scenario

Profit in the PU case is

$$\Pi_u(\lambda, \beta, q) = R_u(\lambda, \beta) - \lambda \frac{s}{q}.$$

The seller's first-order condition is

$$\frac{\partial}{\partial \lambda} R_u(\lambda, \beta_{\hat{\lambda}}) \leq \frac{s}{q}, \quad (15)$$

with equality if $\lambda > 0$. Defining

$$\xi(\lambda) = \left. \frac{\partial}{\partial \lambda} R_u(\lambda, \beta_{\hat{\lambda}}) \right|_{\hat{\lambda}=\lambda}, \quad (17)$$

an equilibrium (λ^*, q^*) in the PU scenario must satisfy

$$\xi(\lambda^*) \leq \frac{s}{q^*}, \quad (18)$$

with equality for $\lambda^* > 0$, and $q^* = 1$ whenever $\lambda^* < \bar{\lambda}_c$.

- **Key difference from PO.** The marginal revenue function is now belief-sensitive. Because the seller can secretly under-recruit without depressing bids ex post, incentives to recruit are weaker.
- **Shape.** $\xi(\lambda)$ is continuous, single-peaked, with $\xi(0) = 0$ and $\xi(\lambda) \rightarrow 0$ as $\lambda \rightarrow \infty$.

3.11 Unobservable-participation equilibrium regions

Let $\underline{\lambda}_u(s)$ and $\bar{\lambda}_u(s)$ be the smaller and larger solutions to $\xi(\lambda) = s$. Proposition 2 gives the full characterization:

$$\lambda_1^* = \underline{\lambda}_u(s), \quad q_1^* = 1; \quad \lambda_2^* = \min\{\bar{\lambda}_u(s), \bar{\lambda}_c\}, \quad (P2)$$

with the no-trade equilibrium $\lambda^* = 0$ always present, and with $q_2^* = 1$ if $\bar{\lambda}_u(s) \leq \bar{\lambda}_c$ and $q_2^* < 1$ otherwise.

- **Main result.** The PU scenario *always* admits a no-trade equilibrium.
- **Multiplicity.** When $s < \xi(\lambda_{\text{peak}})$ and $\bar{\lambda}_c > \underline{\lambda}_u(s)$, there are two additional trade equilibria.

- **Profit ranking.** The larger-trade equilibrium is always strictly more profitable.
- **Economic interpretation.** If bidders expect no trade or low trade, they bid conservatively; this weakens recruitment incentives and validates their expectations. Unobservability thus creates self-fulfilling pessimism.

3.12 Profit comparison across the two scenarios

A central comparative statics identity is

$$R'_o(\lambda) = 2\xi(\lambda), \quad (C)$$

so marginal recruitment incentives are everywhere stronger in the PO scenario than in the PU scenario.

The boundary between the regions in which PO or PU is more profitable is described by Proposition 3. There exists a curve $\hat{s}(c)$ such that

$$\xi(\bar{\lambda}_c) \leq \hat{s}(c) \leq R'_o(\bar{\lambda}_c), \quad (P3a)$$

with

$$\hat{s}(0) = 0, \quad \hat{s}(c) = \xi(\lambda_{\text{peak}}) \text{ on } [c^\#, U(\lambda_{\text{peak}})], \quad (P3b)$$

and $\hat{s}(c) = \xi(\bar{\lambda}_c)$ for sufficiently large c .

- **Interpretation.** Below $\hat{s}(c)$, PU is more profitable; above it, PO is more profitable whenever either scenario trades.
- **Economic logic.** When entry cost c is low, the main issue is to induce aggressive bidding, so PO wins. When recruitment cost s is low and entry is fragile, the weaker recruitment incentives of PU reassure bidders and can dominate.

3.13 Welfare

Welfare is net total surplus:

$$W(\lambda, q) = T(\lambda) - \lambda \frac{s}{q} - \lambda c, \quad T(\lambda) = \int_0^1 [1 - e^{-\lambda(1-G(v))}] dv. \quad (W)$$

The welfare-maximizing participation level satisfies

$$q^w = 1, \quad U(\lambda^w) = c + s \quad \text{if } U(0) > c + s, \quad (20)$$

and the key identity is

$$T'(\lambda) = U(\lambda). \quad (21)$$

- **Interpretation.** The marginal social value of an extra expected participant equals the entrant's expected payoff before paying the entry cost.
- **Key inefficiency.** The equilibrium generally does not satisfy simultaneously the seller's recruitment FOC and the welfare FOC, so $\lambda^* \neq \lambda^w$ for almost all (c, s) .

4 Identification and instruments (what makes the estimates credible)

4.1 Baseline identification idea

- The model is entirely theoretical, so the identification question is not about statistical endogeneity but about the *economic logic* that pins down equilibrium.
- The essential source of identification is the different way in which observability changes the seller's marginal return to recruitment.
- In the PO scenario, observed participation directly affects bids, so recruiting more bidders raises the expected price.
- In the PU scenario, realized participation is hidden. The seller can secretly under-recruit, so bidder expectations need not be validated by actual participation. This weakens recruitment incentives and creates multiple fixed points.

4.2 What makes the PO equilibrium credible

- In the PO case, the seller's recruitment choice is disciplined by the fact that bidders will observe participation before bidding.
- If the seller tries to recruit more, bidders respond through more aggressive bids. This gives a clean single-crossing style structure between recruitment benefits and entry constraints.
- The equilibrium refinement for no-trade beliefs rules out off-path beliefs that would support no trade even when the seller would strictly profit from recruiting.

4.3 What makes the PU equilibria credible

- In the PU case, beliefs over participation are central. If bidders expect low participation, they bid low. This makes recruiting more bidders less attractive to the seller, and the low-participation belief can become self-confirming.
- The no-trade equilibrium is especially robust in the PU scenario because with expected $\lambda = 0$ bidders bid zero, making recruitment unprofitable.
- The smaller positive-trade equilibrium is pseudo-unstable, while the larger positive-trade equilibrium is pseudo-stable. This gives an intuitive reason to focus on the larger-trade equilibrium when discussing profitability.

4.4 Identification of welfare distortions

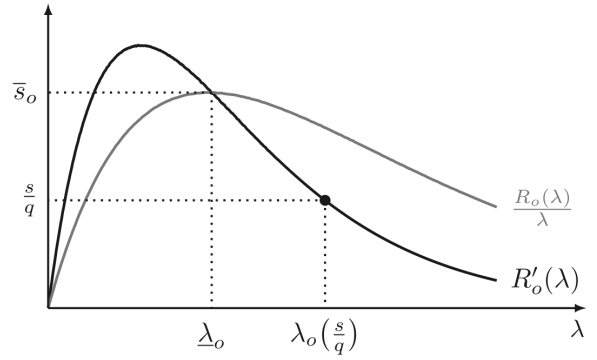
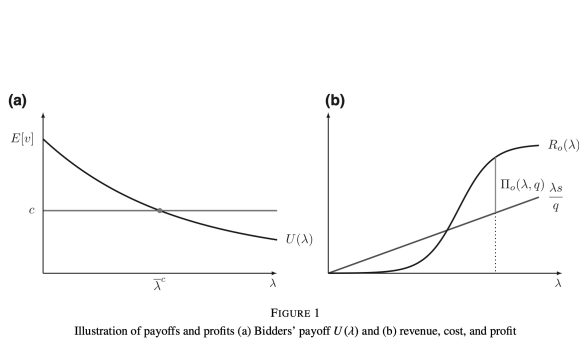
- Welfare differs from equilibrium because the seller maximizes revenue net recruitment cost, whereas the planner values total surplus net recruitment and entry costs.
- The seller internalizes how recruitment changes bids and revenue, but does not internalize all entry-side or total-surplus considerations.
- Therefore, both excessive and insufficient participation are possible in equilibrium, even though entry decisions are conditionally efficient given any fixed λ .

5 Tables and figures

5.1 Figure 1: Illustration of bidder payoffs and seller profits

Read:

- Panel (a) shows that bidder entry value $U(\lambda)$ is strictly decreasing in participation. More competition reduces expected private surplus.
- Panel (b) shows seller revenue and profit against recruitment. Revenue is hump-shaped in average terms, which already suggests interior recruitment choices.
- This figure sets up the basic trade-off of the paper: bidders want lower λ , while the seller often wants higher λ .



5.2 Figure 2: Marginal revenue, average revenue, and marginal recruitment cost

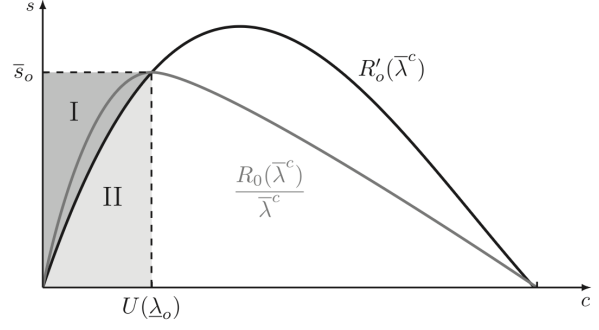
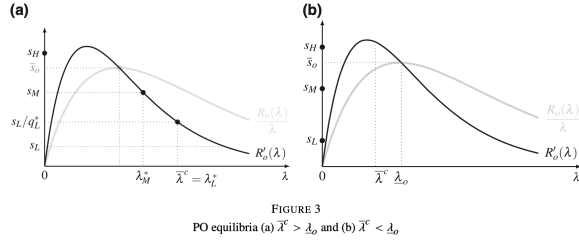
Read:

- The seller's average revenue and marginal revenue are both hump-shaped in λ .
- The horizontal line for s/q shows how bidder reluctance to enter raises the seller's effective marginal recruitment cost.
- The figure defines $\bar{s}_o$ and λ_o , which together determine whether PO trade can be sustained.

5.3 Figure 3: PO equilibria in (λ, s) space

Read:

- Panel (a) shows the two types of positive-trade PO equilibria: unconstrained trade with $q^* = 1$, and entry-constrained trade with $q^* < 1$.
- Panel (b) shows the unraveling case where bidders will not tolerate the seller's minimum viable recruitment scale, so only no trade remains.
- The figure makes visible the paper's first key inefficiency: even profitable trade can disappear because the seller cannot commit to sufficiently low recruitment.



5.4 Figure 4: Unique PO equilibrium with trade regions in (c, s) space

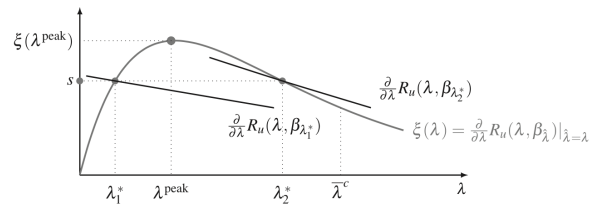
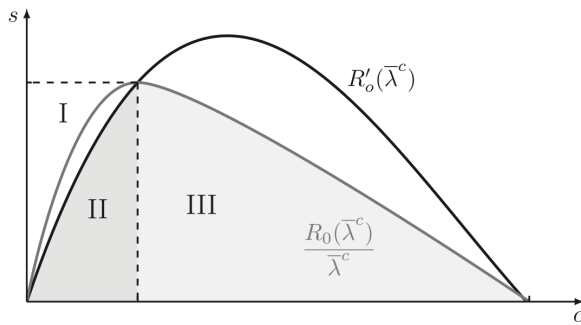
Read:

- Trade occurs only in the shaded Regions I and II.
- Region I is the unconstrained case where bidders are willing to enter at the seller's preferred participation level.
- Region II is the entry-constrained case where the seller must over-recruit and bidders respond with $q^* < 1$.
- The unshaded region is no trade: either recruitment is too expensive or the seller cannot credibly keep participation low enough.

5.5 Figure 5: Value of commitment to recruitment

Read:

- Commitment is strictly profitable only in Regions II and III.
- In Region II, commitment eliminates wasteful over-recruitment needed to offset low entry probability.
- In Region III, commitment rescues trade that otherwise shuts down.



5.6 Figure 6: The function $\xi(\lambda)$ and PU marginal revenue

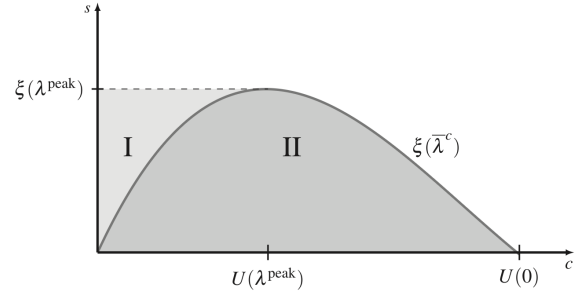
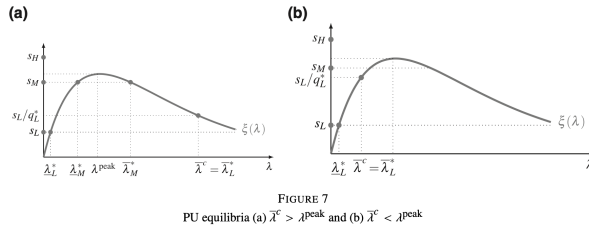
Read:

- The figure introduces the central PU object $\xi(\lambda)$, the marginal revenue of recruitment when beliefs are correct.
- The hump shape of ξ is what creates two possible positive-trade fixed points when s is low enough.
- The horizontal line at s intersecting ξ twice foreshadows multiplicity.

5.7 Figure 7: PU equilibria in (λ, s) space

Read:

- For many (c, s) values, there are three equilibria: no trade, a smaller trade equilibrium, and a larger trade equilibrium.
- The no-trade equilibrium is always present. This is the paper's starkest difference relative to PO.
- The larger positive-trade equilibrium is the more profitable one and is often the natural benchmark for comparing PU to PO.



5.8 Figure 8: PU trade region in (c, s) space

Read:

- Trade occurs only in the shaded region under the curve.
- Region I corresponds to unconstrained PU trade; Region II corresponds to the larger equilibrium being capped by bidder willingness to enter.
- The unshaded area is where only the no-trade equilibrium survives.

5.9 Figure 9: Cases where PO is more profitable than PU

Read:

- These are the low-entry-cost cases. Bidders are willing to enter, so the main issue is how to extract more aggressive bids.

- In panel (a), PU unravels all the way to no trade while PO still trades.
- In panel (b), both trade, but PO recruits more and earns higher profit because observability rewards recruitment with stronger bids.

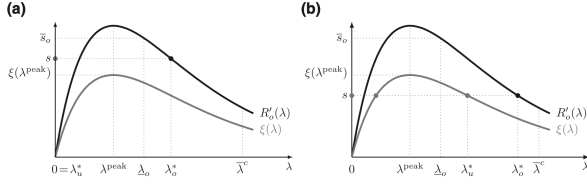


FIGURE 9
Comparison of the PO and PU scenarios when c is relatively small (a) $\xi(\lambda^{\text{peak}}) < s < \bar{s}_o$ and (b) $R'(\bar{\lambda}^c) < s < \xi(\lambda^{\text{peak}})$

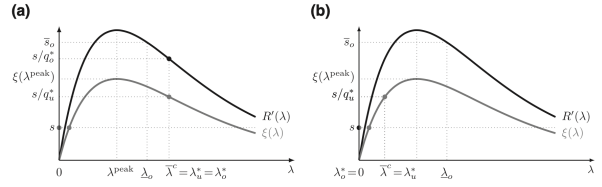


FIGURE 10
Comparison of the PO and PU scenarios when c is relatively large (a) $\bar{\lambda}^c > \Delta_o$ and (b) $\bar{\lambda}^c < \Delta_o$

5.10 Figure 10: Cases where PU is more profitable than PO

Read:

- These are the small- s cases. Recruitment is cheap, so the seller's main problem is to keep bidders from fearing excessive competition.
- In panel (a), both scenarios end up at the same participation level, but PO wastes more recruitment effort because bidders must be more reluctant to enter.
- In panel (b), PO cannot sustain trade at all, while PU can, precisely because weaker recruitment incentives make low expected participation credible.

5.11 Figure 11: Profit ranking of PO and PU across all (c, s)

Read:

- Region I: only PO trades.
- Region II: both trade, but PO is better.
- Region III: both trade, but PU is better.
- Region IV: only PU trades.
- Unshaded: no trade in either scenario.
- This figure is the global summary of the paper's comparative statics.

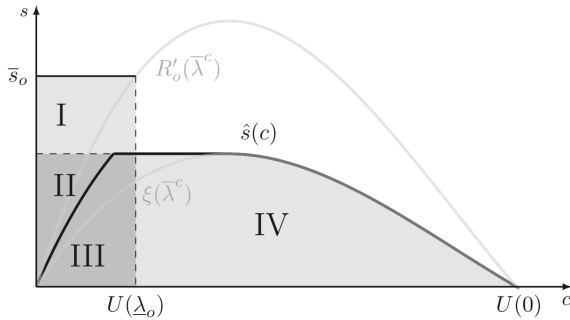


FIGURE 11
Profit ranking of the PO and PU scenarios

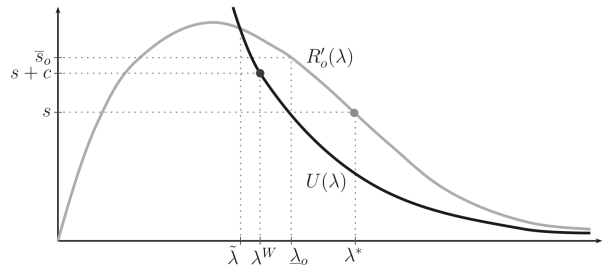


FIGURE 12
Welfare

5.12 Figure 12: Welfare

Read:

- The figure compares the welfare-relevant marginal surplus $U(\lambda)$ to the seller's marginal revenue $R'_o(\lambda)$.
- The planner sets $q^w = 1$ and chooses λ^w from $U(\lambda) = c + s$.
- Because the equilibrium seller solves a different first-order condition, the equilibrium generally misses the welfare optimum.
- The figure makes clear that both under-participation and over-participation are possible.

6 Short summary

- **Conclusion.** Costly recruitment, costly entry, and limited commitment fundamentally alter auction outcomes: trade may unravel, recruitment may be excessive, and the observability of participation can flip which auction environment is more profitable.
- **Intuition.** Observability strengthens recruitment incentives because more participation raises bids. Unobservability weakens recruitment incentives because the seller can secretly under-recruit without lowering bids ex post.
- **Empirical lesson for theory.** In informal auction environments, the main problem is not just auction format but the interaction of participation costs and limited commitment.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- **Main finding.** The paper shows that the interaction of recruitment costs, entry costs, and limited commitment generates equilibrium distortions that are absent when these frictions are studied separately.
- **Equilibrium structure.** The PO scenario has a unique equilibrium except on knife-edge boundaries, while the PU scenario always has a no-trade equilibrium and can have two additional trade equilibria.
- **Comparative ranking.** Neither observability nor unobservability is uniformly better. Which one dominates depends jointly on entry cost c and recruitment cost s .
- **Connection to auction formats.** With unobservable participation, FPA and SPA are no longer equivalent, because they induce different recruitment incentives even though revenue equivalence holds at any given correctly anticipated participation level.
- **Welfare implication.** Equilibrium generally does not maximize welfare. There can be too little trade, too much recruitment, and too much or too little participation.
- **Policy implication.** Commitment devices, disclosure rules, or intermediaries that soften recruitment incentives or coordinate participation can improve outcomes.

7.2 Economic intuition

- The seller wants more bidders for two reasons: to improve the pool of values and to raise the expected price.
- Bidders dislike paying to enter an auction that turns out to be too competitive. This creates a natural congestion externality on participation.
- Observability strengthens the seller’s recruitment motive because realized competition directly disciplines bidding. That helps when the main problem is weak bidding, but hurts when the main problem is bidder reluctance to enter.
- Unobservability weakens recruitment incentives. That can be harmful because it sustains no-trade equilibria, but it can also be beneficial because it creates a form of effective commitment to modest participation.
- The paper’s deepest message is therefore about *endogenous commitment*: what the seller can and cannot credibly promise changes the whole geometry of the participation problem.

7.3 Takeaway

This paper is a clean and powerful example of how realistic trading frictions change not only comparative statics but the very *set of equilibria*. The striking result is that hidden competition can either kill a market or save it, depending on whether the main bottleneck is aggressive bidding or bidder entry. For reading informal auctions, merger negotiations, procurement, or platform design, the right lesson is not simply “more transparency is better” or “more bidders are better” — it is that recruitment incentives, entry costs, and commitment power must be analysed together.